PEARSON EDEXCEL INTERNATIONAL GCSE (9–1) · PHYSICS · 4PH1/1P

A* Examination Guide

Built from 16 question papers and 17 mark schemes, January 2019 to June 2026

This guide is built from the Paper 1P question papers and mark schemes. Model answers follow the marking points examiners list, and every warning comes from an "allow", "ignore", "reject", "condone" or "−1 for POT error" note in the mark schemes. Paper 1 tests the core (non-bold) content of all eight topics; learn the equations, the practical methods and the standard explanations.

1 How Paper 1P is built
2 Twenty mark-scheme rules
3 Equations to recall
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Papers analysed: Jan 2019 (legacy 4PH0), Jun 2019 (mark scheme only), Jan 2020, Nov 2020, Jan 2021, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 25).

1 How Paper 1P is built

Format: 2 hours, 110 marks (the legacy 4PH0 paper was 120), 9–11 structured questions; about 1 minute per mark. Every paper mixes multiple-choice items, "state the equation linking …" (1 mark), calculations (substitution, rearrangement, evaluation), a practical method (4–6 marks), a graph to plot, and 3–6 mark explanations. Recall equations are not given; a few others are printed in the question when needed.

Topic	Papers (of 17)	What is asked most
Forces and motion	17	v–t graphs (gradient, area), stopping distance, F = ma, terminal velocity, Hooke's law
Electricity	17	V = IR, P = IV, E = IVt, I–V circuits, fuses/earthing, thermistor circuits
Radioactivity and particles	16	half-life from a graph, background count, α/β/γ, fission chain, safety
Astrophysics	15	star life cycle, colour and temperature, orbits, orbital speed 2πr/T
Waves (EM and sound)	15	v = fλ, uses and dangers of EM waves, reflection/refraction, sound vs light
Energy transfers and resources	13	KE, GPE, work, efficiency, Sankey diagrams, conduction/convection
Magnetism and electromagnetism	13	field plotting, magnetic materials, motor effect, induced voltage
Solids, liquids and gases	11	density, pressure (F/A, hρg), particle model, gas pressure and temperature

2 TWENTY MARK-SCHEME RULES

1. **Calculations have three marks:** substitution, rearrangement, evaluation. Write the equation, substitute with units converted, rearrange, then give the answer with its unit.
2. **Power-of-ten errors cost one mark** ("−1 for POT error"): convert mA, kW, MHz, GHz, km, minutes and hours first.
3. **Significant figures when asked:** "correct answers not given to 2 s.f. gain 2 marks only" (8.5 s, not 8.55 s).
4. **Equations in words or standard symbols are accepted;** "c" or "C" for current, and "gravity" for g, are ignored or rejected.
5. **g = 10 N/kg is expected;** 9.8 and 9.81 are allowed.
6. **Force labels:** "weight" or "gravitational force", not just "gravity"; "drag" or "air resistance".
7. **Half-life:** "time taken for half the (unstable) nuclei to decay" or "for the activity to halve". "Half the time" is rejected.
8. **Background radiation:** name sources (rocks, radon, cosmic rays, food); CMBR is rejected. Measure it several times, take a mean, subtract it.
9. **Nuclei, not atoms, in nuclear processes** (atoms is condoned once; "daughter cells" is ignored).
10. **Harm needs a reason:** "dangerous" or "harmful" alone is insufficient; say ionising, cell damage, cancer, burns.
11. **Comparisons need both sides:** "comet's orbit is more elliptical than the planet's"; one-sided statements score 0.
12. **Graphs:** linear scales using more than half the grid (no multiples of 3), axes labelled with quantity and unit, points within ½ small square, a smooth curve or straight line of best fit.
13. **Bar charts for categoric data,** line graphs for continuous data ("data are continuous" is a creditworthy reason).
14. **Means:** exclude the anomaly, then round to the precision of the data (19.01 g).
15. **Tick-box rows:** more than one tick in a row scores 0 for that row.
16. **Circuits:** ammeter in series, voltmeter in parallel with the component only, a means of varying the current (variable resistor or power supply), correct symbols.
17. **Methods:** what you measure, with which instrument, what you change, what you keep constant, repeats and means.
18. **Explanations of trends:** describe and then link cause and effect (resistance decreases, so current increases).
19. **"Allow RA":** reverse arguments are credited, so either side of a comparison may be stated, but it must be explicit.
20. **Bald answers are ignored:** "the weather", "bigger magnet", "more energy" without the quantity or mechanism.

3 EQUATIONS TO RECALL

Quantity	Equation	Quantity	Equation
average speed	distance ÷ time	acceleration	$(v - u) \div t$
force	$F = m \times a$	weight	$W = m \times g$
work done	$W = F \times d$	power	$P = W \div t$
kinetic energy	$KE = \frac{1}{2} m v^2$	gravitational PE	$GPE = m \times g \times h$
efficiency	useful ÷ total (× 100%)	density	$\rho = m \div V$
pressure	$p = F \div A$	pressure difference	$p = h \times \rho \times g$

Quantity	Equation	Quantity	Equation
voltage	$V = I \times R$	charge	$Q = I \times t$
electrical power	$P = I \times V$	energy transferred	$E = I \times V \times t$
wave speed	$v = f \times \lambda$	frequency	$f = 1 \div T$

Given in the question when needed: $v^2 = u^2 + 2as$, orbital speed = $2\pi r/T$, $p_1/T_1 = p_2/T_2$ (T in kelvin = °C + 273), and energy = power × time.

4 FORCES AND MOTION

VELOCITY–TIME AND DISTANCE–TIME GRAPHS

Acceleration = gradient of a v–t graph; distance = area under it. Speed = gradient of a d–t graph; horizontal = stationary.

Nov 20: a = −45/39 = −1.2 m/s^2; distance = ½ × 45 × 39 = 880 m. **Jan 22:** F = ma ⇒ a = 7600/1200 = 6.3 m/s^2; $v^2 = u^2 + 2as$ ⇒ s = $21^2 \div (2 \times 6.3)$ = 35 m.

STOPPING DISTANCE (JUN 19, JUN 21, NOV 25)

Stopping distance = thinking distance + braking distance (13.5 + 23.25 = 37 m).

Thinking distance (reaction time × speed): tiredness, alcohol or drugs, distraction (phone), age. Braking distance: speed, mass of car, condition of brakes, tyres and road (wet, icy), slope. A slower reaction increases thinking distance only.

MODEL ANSWER · FALLING OBJECT REACHING TERMINAL VELOCITY (4–6) · JAN 22

At first weight is greater than drag, so the object accelerates. As speed increases, drag increases, so the resultant force and the acceleration decrease (gradient falls). When drag equals weight the resultant force is zero and the object moves at terminal velocity.

- Hooke's law: extension is directly proportional to force up to the limit of proportionality. Elastic: returns to its original shape when the load is removed. A curved section shows the law no longer applies.
- Balanced forces at constant velocity: drag = thrust, weight = lift (draw equal-length arrows).
- Scalar vs vector: a vector has magnitude and direction (velocity, displacement, acceleration, force, momentum).
- Speed conversions (Jan 21): 26 m/s × 3.6 = 94 km/h.

5 ELECTRICITY

WORKED CALCULATIONS

P = IV: 2200 W ÷ 230 V = 9.6 A (choose the 13 A fuse). 350 W ÷ 230 V = 1.5 A.

Potential divider (Jun 25): voltages add to 6.0 V, so the resistor has 2.1 V; I = 2.1/290 = 0.0072 A; $R_{thermistor}$ = 3.9/0.0072 = 540 Ω.

E = IVt (Jan 21): 33 V × 0.085 A × 9000 s × 7 bulbs = 180 000 J. Convert hours to seconds first.

Resistance from an I–V graph (Jun 23): R = V/I = 7.2/2.40 = 3.0 Ω at that point.

MODEL ANSWERS

Why a wire heats up (2–3): the wire has resistance; electrons collide with the lattice ions, transferring energy, so the ions vibrate more and the temperature rises.

Fuse (3): if the current exceeds the fuse rating, the fuse wire melts, breaking the circuit and isolating the appliance. A circuit breaker is faster and can be reset.

Earth wire (2): in a fault (live touching the metal case) a large current flows to earth through the low-resistance wire, which blows the fuse or trips the circuit breaker, preventing a shock.

Thermistor circuit (3): temperature increases, thermistor resistance decreases, current increases, so the voltage across the fixed resistor increases.

- I–V circuits: power supply, variable resistor, component, ammeter in series, voltmeter in parallel. Filament lamp: current increases at a decreasing rate (resistance rises with temperature). Diode/LED: current only above a small forward voltage in one direction.
- Series vs parallel lighting (Jan 21): in series one failed bulb turns all off; parallel bulbs get the full voltage, so each is brighter and can be switched separately.
- Current is the rate of flow of charge; the charge carriers in metals are electrons.
- Live wire is brown and carries the voltage; double insulation means no earth wire is needed.

6 WAVES AND ENERGY

WAVE CALCULATIONS

$v = f\lambda$: $\lambda = 3.0 \times 10^8 \div 5.2 \times 10^9 = 0.058$ m (Jan 22); $\lambda = 3.0 \times 10^8 \div 170 \times 10^6 = 1.8$ m (Jun 23).

Echo (Jun 22): time $= 2 \times 4.70 \div 345 = 0.0136$ s — double the distance for an echo.

Light-time: $t = 1.5 \div 3.0 \times 10^8 = 5.0 \times 10^{-9}$ s (Jun 24).

EM wave	Use	Danger
radio	broadcasting, communications	—
microwave	cooking, satellite, phones	internal heating of body tissue
infrared	heaters, remote controls, thermal imaging	skin burns
visible	fibre optics, photography	—
ultraviolet	fluorescent lamps, security marking	skin damage, cancer, eye damage
X-rays	medical imaging, airport security	ionising: cell mutation, cancer
gamma	sterilising food and equipment, cancer treatment	ionising: cell mutation, cancer

- Transverse (EM, water): vibrations perpendicular to the direction of energy transfer. Longitudinal (sound): parallel.
- Sound vs light (Jun 22): sound is longitudinal and mechanical, needs a medium, is much slower and speeds up in denser media; light is transverse, electromagnetic and travels through a vacuum.
- All EM waves: same speed in a vacuum, transverse, transfer energy, can be reflected, refracted, diffracted.
- Reflection: angle of incidence = angle of reflection, both measured from the normal (drawn at 90°). Refraction practical: ray box, protractor, ruler, glass block; draw the normal in air and glass.
- Energy: $KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 85 \times 4.5^2 = 860$ J; rising height $h = 860 \div (85 \times 10) = 1.0$ m (Nov 24). Work = Fd: 41 000 J ÷ 15 m = 2700 N (Jan 23).
- Sankey diagrams: arrow widths to scale; wasted output usually thermal to the surroundings. Energy is conserved, never destroyed.
- Conduction in metals: vibrating ions pass on vibrations and free electrons carry energy. Convection (4): water near the heater is heated, expands, becomes less dense and rises; cooler denser water sinks; the cycle

repeats.

- Black surfaces are better absorbers and emitters of infrared than white or shiny ones.

7 SOLIDS, LIQUIDS AND GASES

- Density (Jan 19): the same material has the same density whatever the shape, because mass and volume change together.
- Pressure on a surface: $p = F/A$; counting squares to find the area of a foot (Jan 20); two feet halve the pressure.
- Pressure under water (Jan 20): $p = h\rho g = 15.8 \times 1030 \times 10 = 162\,740$ Pa; add atmospheric 100 kPa ⇒ 260 kPa.
- Gas at constant volume (Jun 26): $p_1/T_1 = p_2/T_2$ with T in kelvin: $9.3 \times 755 \div 726 = 9.7$ MPa.
- Particle speed from KE (Jun 26): $v = \sqrt{(2KE/m)} = \sqrt{(2 \times 6.1 \times 10^{-21} \div 7.3 \times 10^{-26})} = 410$ m/s.
- Gas pressure: particles collide with the walls; higher temperature means faster particles, more frequent and harder collisions, so greater pressure.

8 MAGNETISM AND ELECTROMAGNETISM

PLOTTING A FIELD (3)

Place a plotting compass near the magnet, mark the direction of the needle, move the compass and repeat, then join the dots into lines with arrows from N to S. Or sprinkle iron filings on card over the magnet and tap the card.

MODEL ANSWERS

Motor effect (4–6): current in the coil creates a magnetic field; it interacts with the field of the permanent magnets; this produces a force on the wire; forces on opposite sides of the coil are in opposite directions, so it turns; the commutator reverses the current every half turn.

Induced voltage (3): the magnet moves into the coil, the coil cuts the magnetic field lines, a voltage is induced. Increase it by moving faster, using a stronger magnet or more turns.

- Hard magnetic materials (steel) keep magnetism and are hard to demagnetise; soft (iron) magnetise and demagnetise easily.
- Uniform field between flat poles: straight, parallel, evenly spaced lines.
- Like poles repel, unlike attract; field lines never cross.

9 RADIOACTIVITY AND PARTICLES

Radiation	Nature	Penetration	Ionising
alpha (α)	helium nucleus (2p + 2n)	stopped by paper/skin	most
beta (β^-)	fast electron from the nucleus	stopped by a few mm of aluminium	medium
gamma (γ)	high-frequency EM wave	reduced by thick lead/concrete	least

- Beta decay: a neutron becomes a proton; the mass number stays the same, the atomic number increases by 1. Alpha: mass number −4, atomic number −2. Isotopes: same protons, different neutrons.
- Half-life from a graph: draw lines to halve the count (2.6 minutes, Jan 22); after 3 half-lives, $3 \times 5.6 = 16.8$ years.
- Measuring: GM tube and counter; repeat background readings, take a mean and subtract.
- Gamma sterilising fruit (Jun 19): gamma penetrates the box; the fruit does not become radioactive.

- Safety: tongs, lead box, minimise time, maximise distance, never point at people, shielding.

MODEL ANSWER · FISSION CHAIN REACTION (4) · JUN 21

A slow neutron is absorbed by a uranium-235 nucleus, which splits into two daughter nuclei, releasing two or three neutrons and energy; the released neutrons are absorbed by other uranium nuclei, causing further fission.

Reactor: graphite moderator slows neutrons; boron control rods absorb neutrons; thick concrete shielding absorbs radiation.

Fusion (stars) joins light nuclei (hydrogen); fission splits heavy nuclei. Fusion needs high temperature and pressure because the positive nuclei repel (Jun 26).

10 ASTROPHYSICS

LIFE CYCLE OF STARS

Sun-like star: nebula → protostar → main sequence → red giant → white dwarf → black dwarf.

Massive star: main sequence → red supergiant → supernova → neutron star or black hole. The path depends on the star's mass compared with the Sun.

- Colour indicates surface temperature: red (coolest), yellow, white, blue (hottest).
- Orbital speed $v = 2\pi r/T$: $2\pi \times 8.7 \times 10^{10} \div 2.5 \times 10^{8} = 2200$ m/s (Nov 24); convert days or years to seconds first.
- Planets and moons have nearly circular orbits at constant speed; comets have very elliptical orbits, moving fastest near the star, which is at a focus, not the centre.
- Gravity provides the centripetal force (arrow towards the centre); g depends on the mass of the planet or moon.
- Universe > galaxy (Milky Way) > solar system.

11 PRACTICAL SKILLS AND GRAPHS

- Name instruments: metre rule, stopwatch, protractor, ray box, GM tube, newtonmeter, thermometer, balance.
- Method answers: independent and dependent variables, control variables (height, surface, ball), repeats and means; start timing at release and stop on impact.
- Accuracy vs precision: the closeness to the true value vs the spread of repeated readings.
- Plot: linear scales, labelled axes with units, points within ½ small square, curve of best fit with points on both sides; circle the anomaly.
- Hooke's law practical: measure the extension with a ruler and marker, add masses in steps, wear eye protection in case the wire or spring snaps (Jun 26).

12 FINAL A* CHECKLIST

1. Write the equation, convert to SI units (mA, kW, MHz, km, min, h), substitute, rearrange, evaluate, give the unit and the s.f. asked.
2. Use precise terms: weight, drag, nuclei, half the nuclei decay, lattice ions, field lines cut.
3. State harm with a reason (ionising, cell damage, cancer, burns); make comparisons explicit, naming both items.
4. Circuits: ammeter in series, voltmeter in parallel, a variable resistor.
5. Graphs: scales, labels, plots, best-fit curve.
6. Methods: measure, change, control, repeat, mean, safety. Leave 10 minutes to check calculations and units.